

scDesignGuard: source-bound design gating for multisample single-cell differential analysis

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Abstract

Single-cell workflows can produce numerical results even when source identity, biological replication, target support or model estimability are unresolved. We developed scDesignGuard, a source-bound stage-aware framework that returns PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE before inference. Against a frozen source-bound human consensus in an overlap-excluded benchmark of 15 study families, scDesignGuard cleared 4/4 valid designs, blocked 6/6 hard-invalid designs and contained 5/5 unresolved designs, with no unsafe continuation. In four overlap-excluded end-to-end cases, two reached model validity and two stopped at explicit source-binding or donor-replication gates. Nine controlled challenge perturbations produced the prespecified state and reason, and the frozen software reproduced across environments. Applied to 99,029 pancreatic beta cells from 48 donors, scDesignGuard localized same-cohort type 2 diabetes-associated composition, transcriptional and competitive gene-set-enrichment signals. scDesignGuard separates runnable workflows from evidence-supported analyses.

Introduction

Single-cell workflows can complete and return apparently precise results even when the experimental unit or requested contrast is scientifically unsupported. Computational reviews have documented both the power and limitations of scRNA-seq analysis pipelines.¹ Data integration and atlas construction introduce harmonization and reference-selection challenges.^{2–4} Healthy-reference strategies address the distinct problem of identifying disease-altered states against an appropriate baseline.⁵ Extensible ecosystems make cross-study workflows easier to run.⁶ Yet technical executability does not establish inferential validity. Best-practice guidance treats donors or samples—not cells—as the biological replicates in multisample studies.⁷ When cells are analysed as independent replicates, pseudoreplication can inflate significance and false discoveries.^{8–10}

Moving inference to the biological unit is therefore central to multisample analysis. Donor-aware pseudobulk methods generally provide better-calibrated tests than cell-level analyses.^{8–9,11} Cross-study benchmarks show that integration strategy can alter differential-expression results.¹² Individual-level and multisubject differential-expression models explicitly represent between-subject replication.^{13–14} For abundance endpoints, differential-composition methods use complementary frequentist, Bayesian and neighbourhood-based formulations.^{15–17} Flexible-design and variability-aware models extend composition analysis beyond simple two-group settings.^{18–19} For the design problem addressed here, these approaches begin after a scientifically coherent task has been defined; scDesignGuard separately asks whether the source object, donor map, target semantics and estimand jointly authorize that task.

Methodological work also shows that transformation choices can affect single-cell comparisons.²⁰ Latent multivariate regression extends analysis across multiple conditions.²¹ Neighbourhood-based differential expression, alternative

testing paradigms and differential-detection workflows further broaden multisample comparisons.^{22–24} In parallel, simulation studies show that benchmark conclusions depend on whether data-generating mechanisms preserve realistic count structure and gene correlations.^{25–27} Dataset imbalance can also change apparent integration performance.²⁸ Benchmark platforms organize comparable analytic tasks and evaluation workflows.²⁹ For the design problem addressed here, their common starting point is a declared analysis; scDesignGuard evaluates whether the available evidence supports that declaration.

Reproducible infrastructure solves another part of the problem. FAIR software principles define how computational artefacts should be made findable and reusable.³⁰ Research-object packaging and run-provenance records provide machine-readable evidence chains.^{31–32} Workflow guidance formalizes reproducible computational practice.³³ Workflow registries support discovery and reuse.³⁴ Portable workflow systems support repeatable execution.^{35–36} CELLxGENE and GEO provide persistent dataset identities and repository metadata.^{37–38} BioStudies/ArrayExpress and the Human Cell Atlas broaden repository and atlas coverage.^{39–40} Cell Ontology, OBO Foundry and SSSOM support interoperable target semantics and explicit mapping relations.^{41–43} In the stage-aware framework developed here, these resources supply provenance, portability and semantic evidence, whereas the evaluator separately distinguishes demonstrated invalidity from missing evidence and determines stage-specific authorization.

Here we present scDesignGuard, a source-bound, stage-aware framework that compiles these evidence classes into PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE decisions before inference. We evaluate it in an overlap-excluded benchmark of 15 study families, a separate four-case end-to-end gate evaluation, nine controlled challenge perturbations, multi-repository transport audits and clean-environment software reproduction. We then apply the gated workflow to a 99,029-cell pancreatic beta-cell atlas to examine the biological use of evidence-aware design control.

Results

scDesignGuard converts source-bound evidence into stage-specific authorization

scDesignGuard represents an analysis request as a versioned contract that binds the source object, independent biological unit, condition and pairing structure, target ontology, support rule and estimand (Fig. 1a). The evaluator returns PROCEED when evidence required at the current stage is complete, ABSTAIN for modelled uncertainty, BLOCK for a demonstrated invalidity and NON_EVALUABLE when available evidence cannot resolve the requested analysis. Each output carries ordered reasons and an explicit denominator.

Authorization is sequential (Fig. 1b). Source-scope clearance permits structural-design review; structural clearance permits target-support computation; support clearance permits count aggregation and model-validity assessment. Passing a gate authorizes assessment of the next stage, not biological inference.

GSE144735 illustrates the complete path (Fig. 1c). Its paired contract contained six complete pairs, 3,356 target epithelial cells and 12 donor-condition pseudobulks. The model matrix had rank 7/7 and five residual degrees of freedom. Independent implementations reconciled donor assignment, target-cell counts, library sums, pseudobulk dimensions and model rank before final authorization. No differential effect was estimated in this gate evaluation. Development support, robustness and calibration evidence is shown in Extended Data Fig. 1.

An overlap-excluded benchmark demonstrates concordant state assignment

The final overlap-excluded benchmark comprised 15 study families with frozen source-bound human consensus labels: four valid, six hard-invalid and five structurally unresolved. scDesignGuard returned PROCEED for all four valid designs, BLOCK for all six hard-invalid designs and NON_EVALUABLE for all five unresolved designs. Valid clearance was 4/4 (95% Wilson CI, 0.510–1.000), hard-invalid detection 6/6 (0.610–1.000), unresolved containment 5/5 (0.566–1.000), unsafe continuation 0/6 (0.000–0.390) and exact state concordance 15/15 (0.796–1.000; Fig. 2).

The reference review was blinded to evaluator, executable-checklist, end-to-end and biological-outcome outputs, but verified prepared source-bound fields rather than independently compiling every semantic contract field de novo. These endpoints therefore quantify concordance with frozen source-bound human consensus, not accuracy against an independently constructed external reference.

End-to-end cases show when analyses proceed and when they stop

A separate overlap-excluded end-to-end cohort comprised four cases processed in frozen order through source scope, structural design, target support, count aggregation, model validity and final authorization (Fig. 3a,c).

GSE144735 contained six complete pairs, 3,356 target cells, 12 pseudobulks, a full-rank 7-column design and five residual degrees of freedom. GSE123813 contained 11 complete pairs, 33,106 target cells, 22 pseudobulks, a full-rank

12-column design and ten residual degrees of freedom (Fig. 3b). Both reached final authorization. Human Cell Atlas project 135f7f5c-4a85-4bcf-9f7c-4f035ff1e428 stopped because exact donor-condition-modality and raw-count-layer binding could not be established. GSE81076 stopped because five total donor identifiers cannot satisfy the frozen minimum of five donors per condition.

Python primary computation, an independently compiled C count implementation and an independent R design-matrix implementation reconciled donor assignment, target-cell counts, raw-library sums, pseudobulk dimensions, rank, condition-effect estimability and residual degrees of freedom. Differential expression, pathways and biological effects were not computed in this evaluation.

Controlled challenge perturbations and reproducible transport define the operating boundary

Nine deterministic perturbations altered one design property at a time: donor deletion, cell-level pseudoreplication, batch-condition confounding, duplicate donor, pair destruction, wrong target, wrong version, mixed strata or low support. All nine produced a safe non-PROCEED state and the prespecified primary reason (9/9 for both endpoints; Fig. 4a). These perturbations test rule sensitivity rather than estimate defect prevalence.

Repository transport audits covered 18 study families across BioStudies/ArrayExpress, Single Cell Portal, NCBI GEO and Human Cell Atlas (Fig. 4b). These Tier C audits evaluated source identity, access boundaries and terminal reasons without target, expression or outcome analysis; they are not effect-performance cases. The controlled-challenge and repository-transport details are shown in Extended Data Fig. 3.

The frozen wheel, OCI image, JSON Schema, reason registry and terminal-state fixtures reproduced in a clean environment. Three current-release suites and two compatibility suites reproduced their expected behavior, and the repository run yielded 1,362/1,362 passed tests, 29 planned skips and 42 passed subtests (Fig. 4c). Engineering reproducibility is distinct from scientific validation. Synthetic uncertainty and clean-environment reproduction are summarized in Extended Data Fig. 2.

Donor-aware gating localizes same-cohort beta-cell signals

The same-cohort biological application used the GSE221156 pancreatic islet atlas⁴⁴ and comprised 99,029 author-annotated beta cells from 48 donors: 17 normal, 14 prediabetes and 17 type 2 diabetes donors across author-defined clusters C0–C7. Counts were aggregated once per donor–cluster. C7 had the lowest qualifying support (13 normal, eight prediabetes and ten type 2 diabetes donors) and was interpreted cautiously.

Recovered beta-cell-state composition differed between type 2 diabetes and normal donors after adjustment for assay, sex, ethnicity, age and body mass index (Fig. 5a,b). Mean donor C6 proportions were 0.060, 0.086 and 0.183 in normal, prediabetes and type 2 diabetes, respectively; the adjusted type 2 diabetes–normal coefficient was 0.191 on the arcsine-square-root scale (BH-FDR across eight clusters, 1.78×10^{-4}). C0 decreased from 0.242 to 0.137 (coefficient, -0.122 ; FDR, 0.0213). These proportions describe recovered beta cells, not absolute islet abundance. Beta-cell quality control, qualifying-donor support and donor-level composition are shown in Extended Data Fig. 5.

Counts were aggregated at donor level before expression modelling.^{8–9,11} We fitted edgeR quasi-likelihood models to 100,434 eligible gene–cluster entries.⁴⁵ The type 2 diabetes–normal contrast yielded 418 associations involving 225 unique genes (Fig. 5c). Reproduction of the source-atlas whole-beta workflow⁴⁴ yielded 511 genes; 136/225 state-resolved genes overlapped this set and accounted for 310/418 associations, all direction-concordant. Limma-voom sensitivity analysis^{46–47} yielded 444 associations, 311 shared with edgeR and no shared sign conflict. Reactome⁴⁸ and Hallmark⁴⁹ collections were tested competitively with CAMERA.⁵⁰ CAMERA showed significant negative competitive gene-set signals for fatty-acid-metabolism-associated gene sets in all eight clusters and oxidative-phosphorylation-associated gene sets in seven clusters, significant positive competitive gene-set signals for TNF–NF- κ B-associated transcriptional programs in six clusters, and significant negative competitive gene-set signals for mTORC1-signalling-associated gene sets in six clusters (Fig. 5d). These findings localize same-cohort signals rather than establish independent replication. Whole-beta sensitivity analyses and the invalid cell-level counterfactual are shown in Extended Data Fig. 4; differential-expression, competitive gene-set-enrichment and atlas-overlap diagnostics are shown in Extended Data Fig. 6.

Discussion

scDesignGuard formalizes a distinction that computational pipelines can obscure: technical execution does not itself authorize scientific inference. The framework makes source identity, biological replication, target support and estimability explicit prerequisites and records where authorization stops. It complements rather than replaces donor-aware statistical engines.

The 15-family overlap-excluded benchmark shows that the evaluator is not an always-BLOCK or always-NON_EVALUABLE rule set: valid designs proceeded, demonstrated invalidities were blocked and unresolved records remained non-evaluable. The exact state concordance is accompanied by wide Wilson intervals because the denominator is modest. The frozen source-bound human consensus, checklist and evaluator also share a project taxonomy, so these results support deterministic state concordance within that shared taxonomy rather than accuracy against an external criterion or superiority to an independently authored expert method.

The four-case end-to-end evaluation complements the benchmark by showing that PROCEED can reach quantitatively supported target aggregation and an estimable model, whereas prespecified early stops preserve interpretable source-binding or donor-replication reasons. This denominator is also modest, and no differential effect was estimated in the gate evaluation.

Controlled challenge perturbations, repository transport, clean-environment software reproduction and the same-cohort beta-cell application answer different questions. They demonstrate rule sensitivity, portability of structural evidence, deterministic implementation and biological utility, respectively, but none substitutes for the others. The beta-cell analysis localizes donor-aware signals within one cohort and provides neither independent replication nor causal or clinical claims.

Semantic contract construction remains the principal practical limitation because trained human interpretation is still required for complex source records, ontology relations and estimands. Future work should measure fully independent contract-construction reproducibility, time and usability, and should expand overlap-excluded end-to-end evaluation. Within the present boundaries—modest benchmark and end-to-end denominators, shared taxonomy, one-cohort biology and no clinical or causal interpretation—scDesignGuard provides a reproducible layer that separates runnable workflows from evidence-supported inference.

Methods

Framework and scientific states

A task contract binds immutable source identifiers and versions, the independent biological unit, condition and pairing fields, source-defined target labels and ontology relations, support thresholds, estimand definitions and evidence-object references. The evaluator emits PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE with a primary and ordered secondary reason set. Rules are stage-specific: source scope, structural design, target support, count aggregation, model validity and biological inference are authorized separately. ABSTAIN was retained as a prespecified framework state for modelled uncertainty but was not activated by any final benchmark or end-to-end case; its empirical performance is therefore not evaluated here.

Evidence tiers and decision hierarchy

Tier A requires at least five donors per group or five complete pairs, at least 20 target cells per donor, exact donor mapping, EXACT or CHILD target ontology, established analysis access and a complete estimand. Tier B requires at least three donors per group or three complete pairs and at least ten target cells per donor and is limited to supportive robustness or direction. Tier C is restricted to metadata, provenance, repair and terminal-state evaluation. Tier D comprises known-invalid transformations of public real-data contracts. The donor and target-cell thresholds are conservative project-defined authorization cut-offs rather than universal sample-size recommendations. They distinguish full analytical authorization from lower-evidence robustness or structural use.

Terminal state was selected using the frozen precedence order BLOCK, NON_EVALUABLE, ABSTAIN and PROCEED. All activated reason codes were ordered first by terminal-state precedence and then lexicographically by code; the first code was reported as the primary reason and the remaining codes as ordered secondary reasons. In stage-ledger reporting, unperformed or stage-inapplicable checks remained NOT_ASSESSED or NOT_APPLICABLE. Supplementary Table 3 documents the 21-code NM03 v0.1.0 evaluator registry; frozen benchmark-compiler reason labels used for the 15-family benchmark are reported separately in Supplementary Table 1.

Final benchmark selection and human consensus

Two human reviewers verified prepared source-bound records while blinded to evaluator, executable-checklist, end-to-end and biological-outcome outputs; disagreements were resolved and the consensus labels were frozen. The 24 frozen candidates were subsequently audited against a project-wide family-identity registry linking repository accessions or collection identifiers with publication, project and cohort aliases. Nine families with identified development or calibration study-family overlap were excluded without replacement before final endpoint estimation, leaving 15 families. Exclusion was based solely on family identity and project role and did not use evaluator output,

downstream accessibility, biological-effect estimates or result appearance. Because the review verified prepared source-bound fields rather than independently constructing every semantic field de novo, the reference is described as frozen source-bound human consensus rather than an independently constructed criterion standard. Reviewer-to-reviewer inter-rater reliability was not treated as an endpoint.

Benchmark endpoints

VALID_DESIGN, HARD_INVALID_DESIGN and STRUCTURALLY_UNRESOLVED reference classes implied expected PROCEED, BLOCK and NON_EVALUABLE states, respectively. Primary endpoints were valid clearance, hard-invalid detection, unresolved containment, unsafe continuation and exact state concordance. Each rate is reported with exact membership and a two-sided 95% Wilson interval. Evaluator and executable-checklist outputs did not modify the human reference. These endpoints evaluate deterministic state concordance within a shared project taxonomy and are not estimates of superiority over an independently authored expert method.

End-to-end target-support and model-validity evaluation

A separate four-case overlap-excluded cohort was processed in frozen order through source scope, structural design, target support, count aggregation, model validity and final authorization. Candidate inclusion did not use scDesignGuard state, downstream effect estimates or biological outcomes.

Independent comparisons required at least five qualifying donors per condition; paired comparisons required at least five complete pairs. Each donor-target unit required at least 20 target cells and a finite positive raw integer library. Repeated samples were aggregated to donors.^{8–9,11} Model validity required a full-column-rank matrix, an estimable condition effect and positive residual degrees of freedom. A Python implementation computed donor assignment, target-cell counts, raw-library sums, pseudobulk dimensions, rank, estimability and residual degrees of freedom. Count quantities were recomputed by an independently compiled C implementation and model-matrix quantities by an independent R implementation. Differential expression, pathways and effects were not computed.

Controlled challenge perturbations and repository transport

Nine transformations modified one prespecified property while retaining all other contract fields: donor deletion, cell-level pseudoreplication, batch-condition confounding, duplicate donor, pair destruction, wrong target, wrong version, mixed strata and low support. Expected states and primary reasons were fixed before evaluation.

Repository transport used official metadata and identity endpoints for CELLxGENE,³⁷ NCBI GEO³⁸ and BioStudies/ArrayExpress.³⁹ Single Cell Portal and Human Cell Atlas sources were queried through their official metadata interfaces; the Human Cell Atlas context is described elsewhere.⁴⁰ Authentication failures, absent row metadata and unresolved donor mappings were retained as evidence states rather than inferred from descriptive text. Expression and outcomes were not accessed for transport-audit families.

Software reproducibility

The evaluated software was frozen as a Python wheel, OCI image, JSON Schema, 21-code reason registry, four terminal-state fixtures and software bill of materials. Packaging and provenance followed FAIR software principles and computational-workflow guidance.^{30,33} Clean-environment installation, current-release tests, compatibility tests and repository-boundary fixtures were executed against recorded hashes.

Pancreatic beta-cell analysis

The exact CELLxGENE object³⁷ for GSE221156⁴⁴ contained 99,029 author-annotated beta cells from 48 donors. Author C0–C7 labels were retained without marker-based renaming. Raw counts were summed once per donor–cluster. A donor–cluster unit required at least 20 cells and a positive raw library sum.

Composition models used arcsine-square-root donor proportions and adjusted for assay, sex, ethnicity, age and body mass index; inference used limma empirical-Bayes tests⁴⁷ with BH correction across eight clusters per contrast. Counts were aggregated at donor level to preserve the biological unit.^{8–9,11} Cluster-specific models were fitted with edgeR quasi-likelihood methods⁴⁵ using the same covariate design. Genes were filtered within cluster, and BH correction was applied across all eligible gene–cluster tests within each contrast. Limma-voom^{46–47} used identical matrices, covariates and contrasts. All eligible Reactome⁴⁸ and Hallmark⁴⁹ gene sets were tested competitively with CAMERA,⁵⁰ with BH correction across the complete cluster–gene-set universe within each collection and contrast. For Fig. 5d, the 12 Hallmark gene sets with the smallest minimum global cluster–gene-set FDR across C0–C7 in the type 2 diabetes–normal contrast were displayed. Extended Data Fig. 6 applied the same ranking rule separately to display 15 Hallmark and 20 Reactome gene sets. Display selection did not change either multiple-testing universe.

The atlas-overlap analysis reproduced the GSE221156 authors' whole-beta type 2 diabetes–normal workflow,⁴⁴ yielding 511 genes. State-resolved genes and gene–cluster associations were joined to that set in a shared 13,207-gene test universe. Overlap denominators were 225 unique genes and 418 significant associations; direction concordance was evaluated only for overlapping associations.

Statistics and reporting

Cells were never treated as independent biological replicates.^{8–10} Binomial intervals are two-sided Wilson intervals. Differential-expression and composition P values are two-sided and adjusted as specified above. Competitive enrichment direction is encoded by the sign of the CAMERA test statistic. No threshold, endpoint or truth rule for the final 15-family overlap-excluded benchmark or the four-case end-to-end evaluation was modified after those analysis sets and endpoint definitions were frozen.

Data availability

The final 15-family benchmark membership, four-case end-to-end gate results, controlled-challenge results, repository-transport summaries, quantitative reconciliation and figure Source Data accompany the manuscript. Historical development and audit objects are retained under version control but are not part of the submitted scientific denominators. Expression objects are not redistributed and remain subject to repository and study terms. The same-cohort biological application is available under accession GSE221156.

Code availability

The evaluated scDesignGuard v0.1.0 source code and reproducibility materials are publicly available from the GitHub release at <https://github.com/wang177777/scDesignGuard/releases/tag/v0.1.0>. The exact evaluated wheel is contained in the public evaluated-code archive, and the exact evaluated OCI archive is provided as a standalone release asset; both are checksum-bound in the release. The software is distributed under the BSD 3-Clause license. A persistent DOI archive has not yet been established. Third-party expression objects and participant-level repository payloads are not redistributed; their official accession or collection identifiers are provided in the manuscript and Source Data.

Ethics statement

This study exclusively performed secondary analyses of publicly available, de-identified human datasets. No new participants were recruited, no intervention was performed, no new biological specimens were collected, and the investigators had no access to identifiable private information. Accordingly, under applicable institutional policies, additional ethics committee approval and informed consent were not required for this secondary analysis. Ethics approvals and informed-consent or waiver procedures for the original data collections are reported in the respective source publications and repositories. The secondary use and reporting of the data followed applicable repository terms, institutional policies and the relevant ethical principles of the 2024 Declaration of Helsinki.

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Author contributions

Guoyong Wang, Kaijun Zhang, Jiyue Jiang, Yu Li and Xiaonan Yang conceived the study. Kaijun Zhang, Jiyue Jiang and Yu Li developed the methodology. Jiyue Jiang and Yu Li developed the software, curated the data and performed the formal analysis. Guoyong Wang, Weixin Wang, Chaonan Wang, Hui Bi, Haojun Liang, Zuoliang Qi and Ying Huang contributed to clinical and biological interpretation. Guoyong Wang, Kaijun Zhang, Jiyue Jiang and Yu Li drafted the manuscript. All authors reviewed, edited and approved the final manuscript. Guoyong Wang and Xiaonan Yang

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Competing interests

The authors declare no competing interests.

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Figure legends

Figure 1 | Source-bound design evidence and stage-specific authorization. a, scDesignGuard compiles source identity, donor and condition structure, target ontology, support and estimand into an evidence-bearing contract that returns PROCEED, ABSTAIN, BLOCK or NON_EVALUABLE with explicit reasons. b, Sequential gates comprise source scope, structural design, target support, count aggregation, model validity and inference authorization; passing one gate authorizes only assessment of the next gate. c, GSE144735 worked example: six complete pairs, 3,356 target epithelial cells, 12 donor-condition pseudobulks, design-matrix rank 7/7 and five residual degrees of freedom. No inferential test or biological effect was computed in this gate evaluation.

Figure 2 | Overlap-excluded benchmark and state concordance. a, Frozen source-bound human consensus was established before a project-wide family-identity audit defined the final overlap-excluded endpoint set. b, Frozen human-reference strata: four valid, six hard-invalid and five structurally unresolved families. c, Exact proportions and two-sided 95% Wilson intervals for valid clearance (4/4), hard-invalid detection (6/6), unresolved containment (5/5), unsafe continuation (0/6) and exact state concordance (15/15). d, Family-level state matrix. These endpoints quantify concordance with frozen source-bound human consensus, not independently labelled external-accuracy estimates.

Figure 3 | Overlap-excluded end-to-end outcomes. a, Outcomes for four cases. GSE144735 and GSE123813 passed all gates; Human Cell Atlas project 135f7f5c-4a85-4bcf-9f7c-4f035ff1e428 stopped at structural source binding and GSE81076 stopped at the donor-replication threshold. b, Exact quantitative checks for the two full gate passes. GSE144735 comprised six complete pairs, 3,356 target cells, 12 pseudobulks, rank 7/7 and five residual degrees of freedom; GSE123813 comprised 11 complete pairs, 33,106 target cells, 22 pseudobulks, rank 12/12 and ten residual degrees of freedom. c, Number of cases clearing each sequential gate. No differential expression, pathway analysis or biological-effect estimation was performed.

Figure 4 | Controlled challenge perturbations, repository transport and software reproduction. a, Terminal states for nine deterministic challenge perturbations. Five returned BLOCK and four NON_EVALUABLE; all nine returned a safe non-PROCEED state and the prespecified primary reason. These are controlled challenge perturbations, not prevalence-weighted cases. b, Eighteen Tier C repository transport audits across BioStudies/ArrayExpress (n=5), Single Cell Portal (n=4), NCBI GEO (n=5) and Human Cell Atlas (n=4). Expression and outcomes were not accessed. c, Reproduction summaries: 3/3 current-release suites, 2/2 compatibility suites and 1,362/1,362 repository tests passed; 29 tests were skipped by design and 42 subtests passed. No statistical test was performed for b or c.

Figure 5 | Donor-aware beta-cell composition, transcription and competitive gene-set enrichment. a, Mean donor proportions of author-defined clusters C0–C7 in normal (ND; n=17 donors), prediabetes (PD; n=14) and type 2 diabetes (T2D; n=17); bars are arithmetic means across donors. b, Adjusted T2D–ND composition coefficients from donor-level models on the arcsine-square-root scale after adjustment for assay, sex, ethnicity, age and body mass index. Points are coefficients; error bars are descriptive ± 1.96 unmoderated coefficient standard errors (residual d.f.=39). P values are two-sided limma empirical-Bayes tests; FDR values are Benjamini–Hochberg adjusted across eight clusters. Positive values indicate higher recovered-cell proportion in T2D. c, Significant gene–cluster associations after BH correction across 100,434 eligible tests in the T2D–ND contrast. Bars show edgeR quasi-likelihood and limma-voom counts. edgeR identified 418 associations involving 225 genes; 136/225 genes overlapped the same-cohort 511-gene whole-beta atlas set, accounting for 310/418 associations with 310/310 direction agreement. d, Signed $-\log_{10}$ global FDR for competitive Hallmark CAMERA gene-set tests. All eligible Hallmark cluster–gene-set tests were included in BH correction; for visualization, the 12 gene sets with the smallest minimum global FDR across C0–C7 were displayed. Red denotes positive and blue negative competitive gene-set signals in T2D. Display selection did not alter the inferential universe. Results are same-cohort localization, not independent replication.

Extended Data figure legends

Extended Data Figure 1 | Development support, robustness and calibration. a, Qualifying independent-unit support under the development protocol; the dashed line marks the project-defined five-donor cut-off. b, Leave-one-donor-out direction-stability profiles. c, Calibration-label composition. d, Minimum target-cell support among qualifying donor-target units; the dashed line marks the project-defined 20-cell cut-off. These data are development evidence, not benchmark performance.

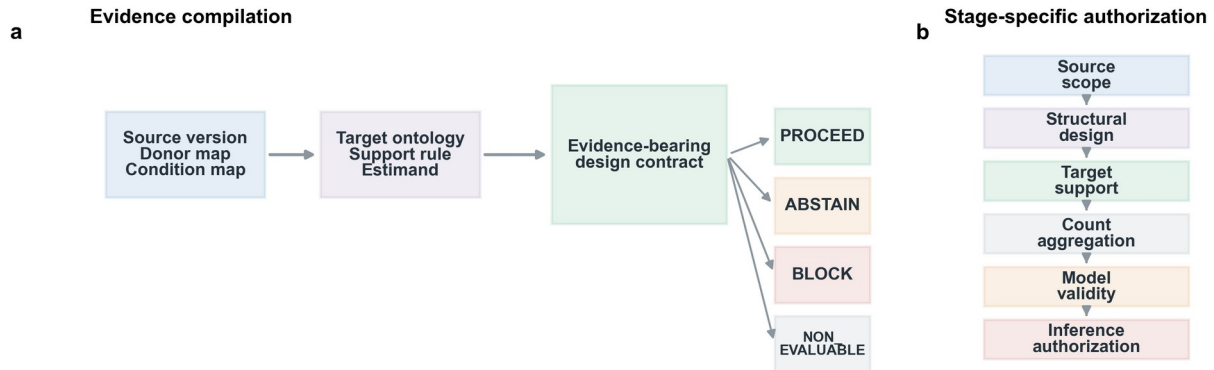
Extended Data Figure 2 | Synthetic uncertainty and software reproduction. a, Descriptive Monte Carlo estimates for pooled Type-I rate and mean replicate FDP/FDR with simulation intervals and Monte Carlo standard errors. The 0.05 line is a reference, not a post hoc claim boundary. b, Public v0.1.0 GitHub release, exact evaluated wheel in the public release archive, standalone exact OCI archive, BSD 3-Clause license, schema and fixture checks, clean-environment installation and security checks. A persistent DOI archive has not yet been established. Synthetic estimates do not support strict nominal Type-I or FDR-control claims.

Extended Data Figure 3 | Controlled challenge perturbations and repository transport. a, Nine controlled challenge perturbations; five returned BLOCK and four NON_EVALUABLE, with observed terminal states and primary reasons. b, Tier C transport-audit families by repository and time stratum. c, Matched clean–challenge contract checks. Expression and outcomes were not accessed for transport audits; no biological effect was estimated.

Extended Data Figure 4 | Whole-beta robustness and invalid counterfactual. a, Leave-one-donor-out effect correlations. b, Direction-retention fractions. c, Filtering sensitivity. d, Invalid cell-level pseudoreplication counterfactual. The counterfactual is a methodological demonstration and carries no biological interpretation.

Extended Data Figure 5 | Beta-cell quality control, support and composition. a, EdgeR multidimensional-scaling coordinates for donor–cluster pseudobulks, coloured by condition; no point was excluded after inspection. b, Qualifying donor counts under the ≥ 20 -cell rule. c, Donor-level recovered-beta-cell composition. Box plots show medians, interquartile ranges and whiskers extending to $1.5\times$ the interquartile range.

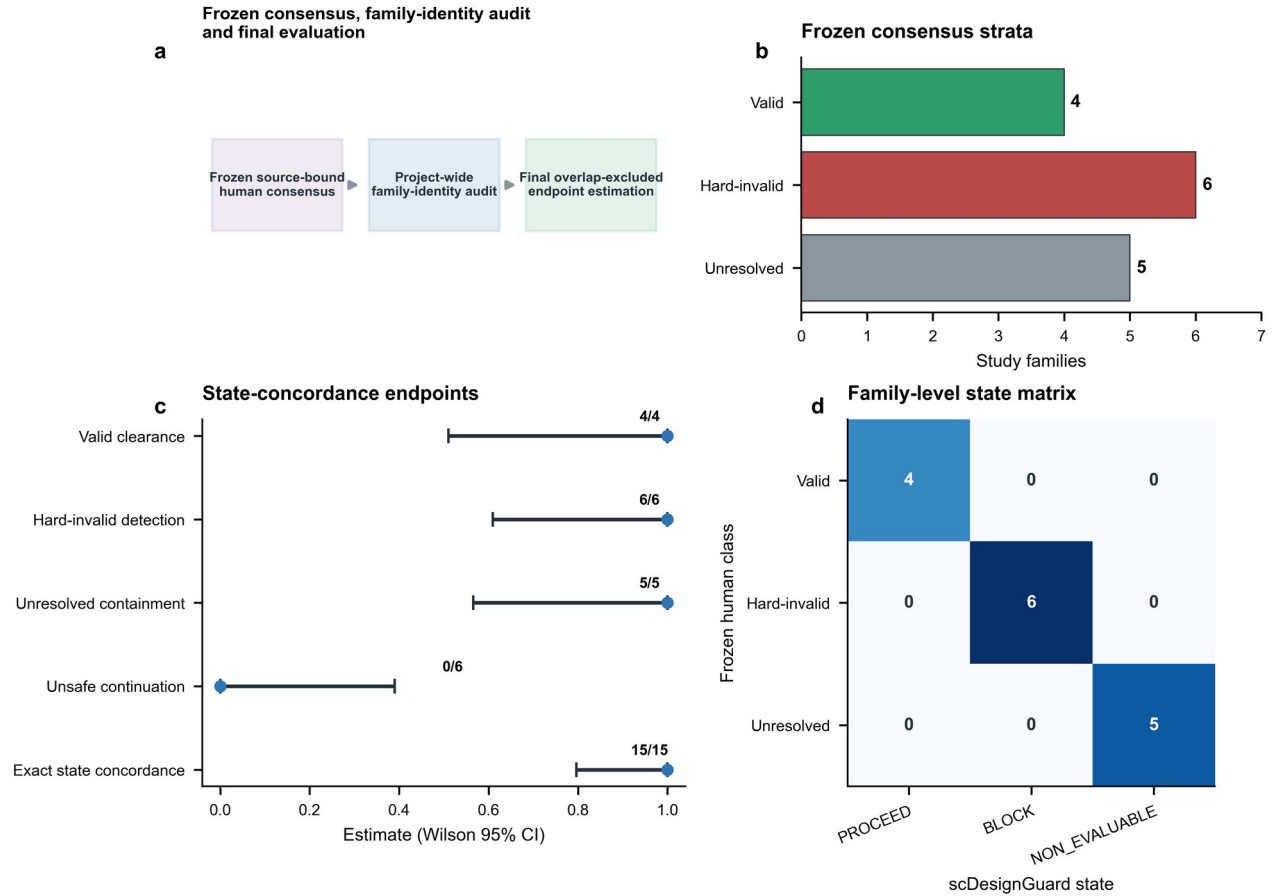
Extended Data Figure 6 | Beta-cell differential expression, competitive gene-set enrichment and atlas overlap. a, EdgeR quasi-likelihood T2D–ND effects; red points pass BH correction across all eligible gene–cluster tests. b, c, Signed $-\log_{10}$ global FDR for Hallmark and Reactome CAMERA gene-set tests. The 15 Hallmark and 20 Reactome gene sets with the smallest minimum global cluster–gene-set FDR across C0–C7 were displayed separately; all eligible tests in each collection remained in the BH correction universe. d, edgeR–voom log-fold-change concordance. e, Same-cohort overlap: 136/225 state-resolved genes and 310/418 associations overlap the whole-beta atlas set, with 310/310 direction concordance. State-resolved-only genes are not automatically novel findings.



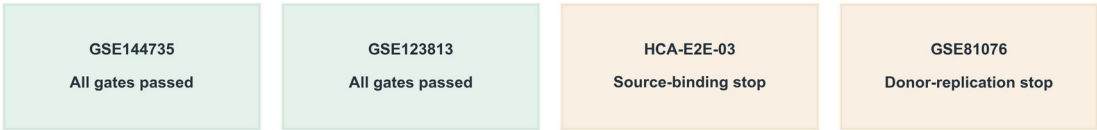
c Worked example: GSE144735



Frozen consensus, family-identity audit and final evaluation

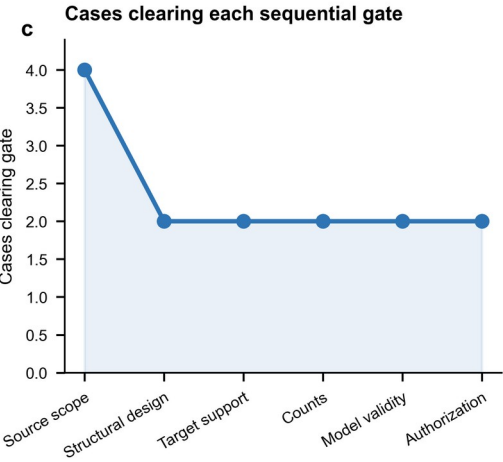


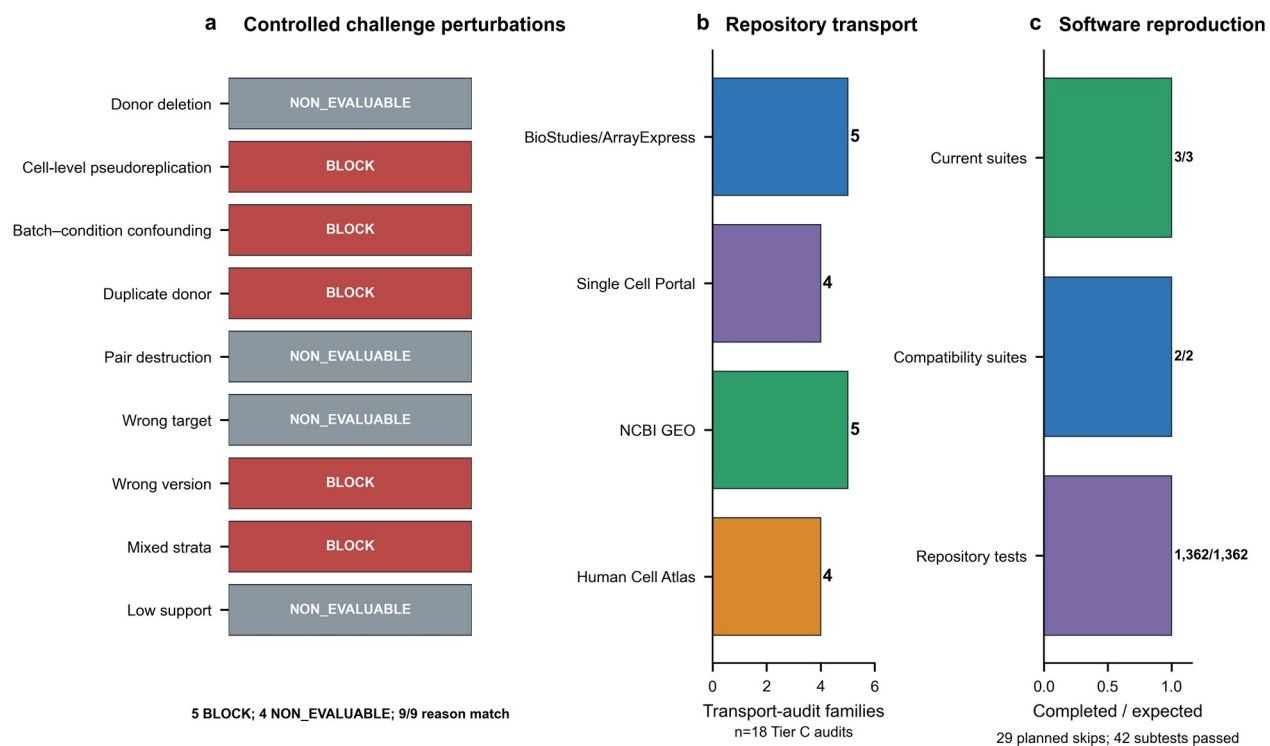
a Final gate outcome

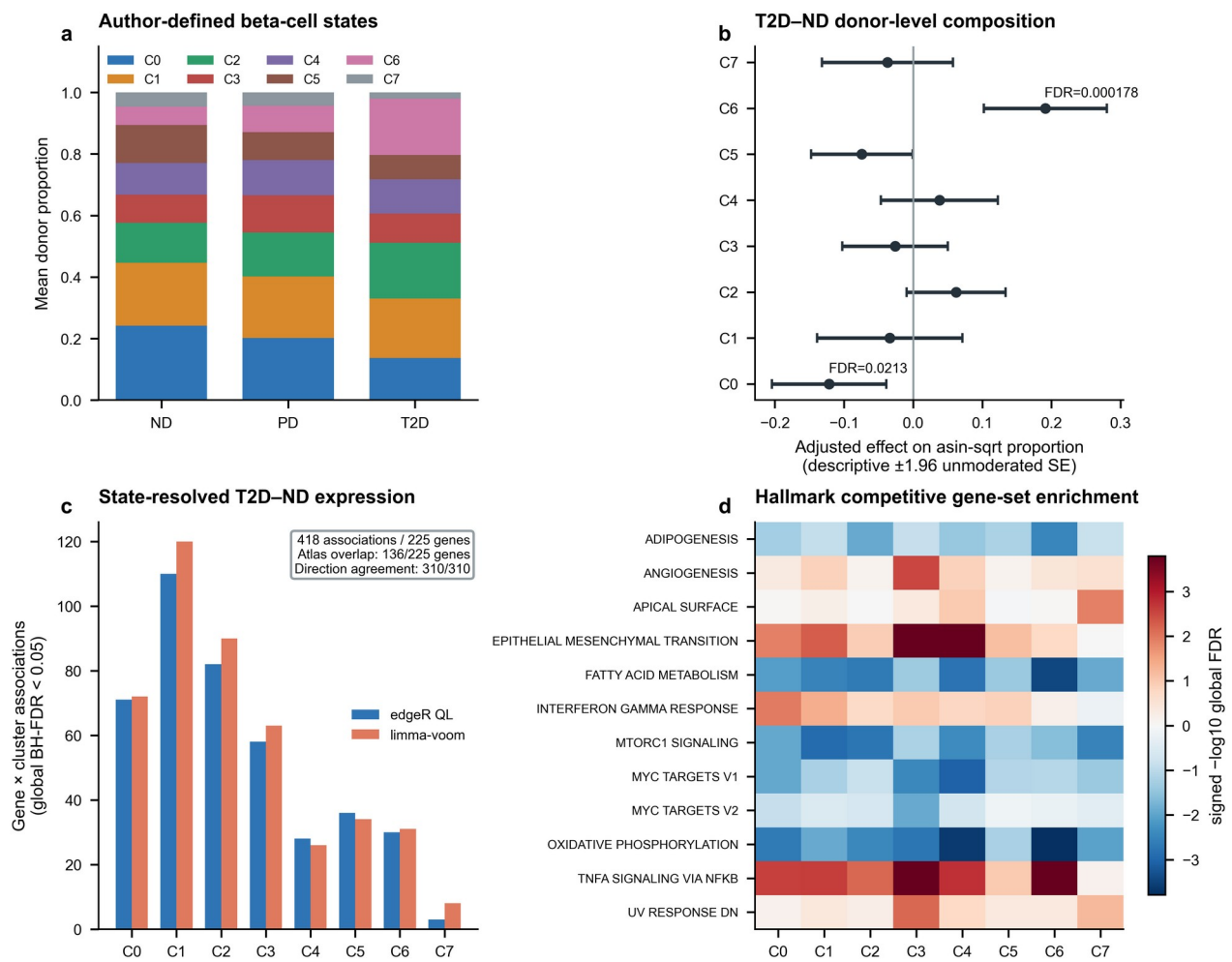


b Quantitative gate checks

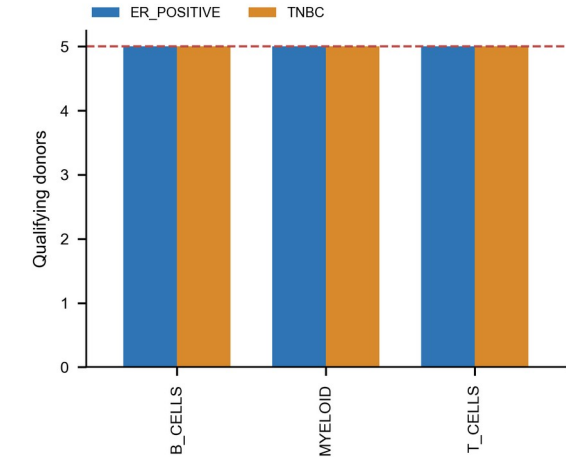
	Complete pairs	Target cells	Pseudobulk rows	Design rank	Residual d.f.
GSE144735	6	3,356	12	7/7	5
GSE123813	11	33,106	22	12/12	10



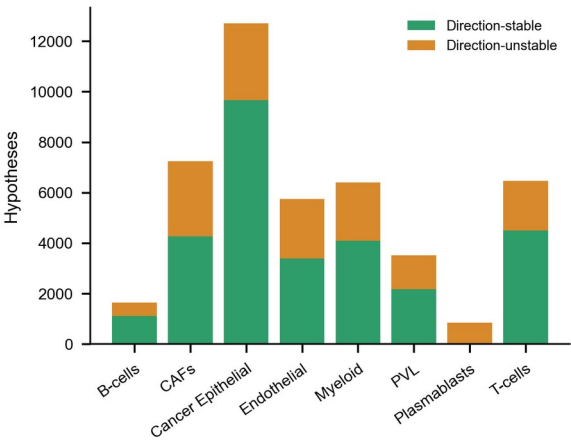




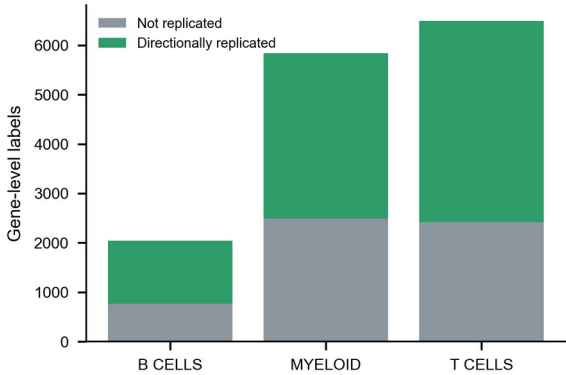
a Independent-unit support



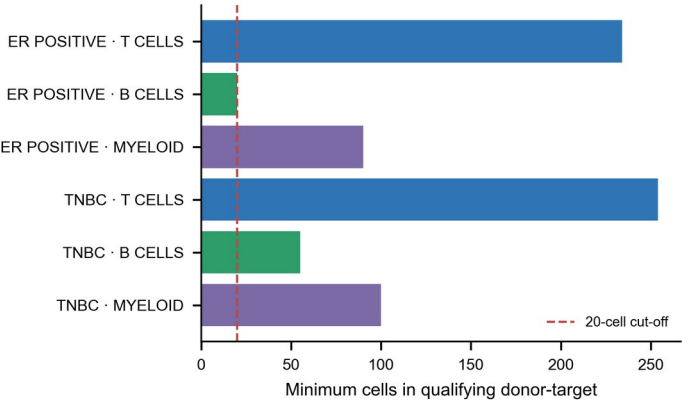
b Leave-one-donor-out diagnostics

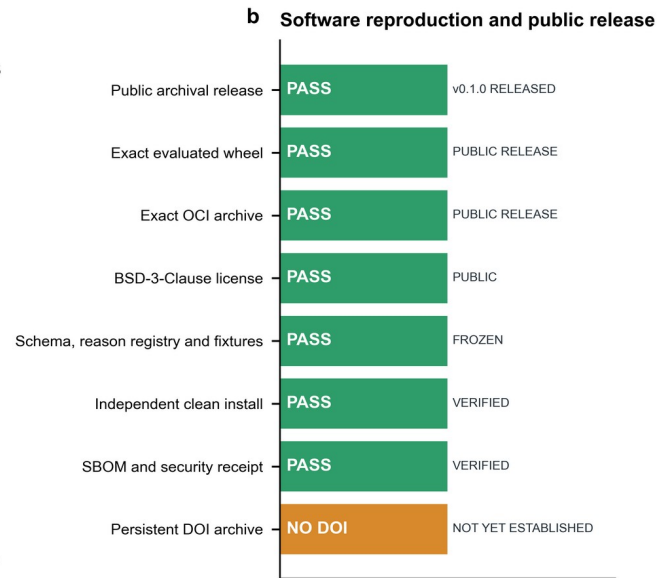
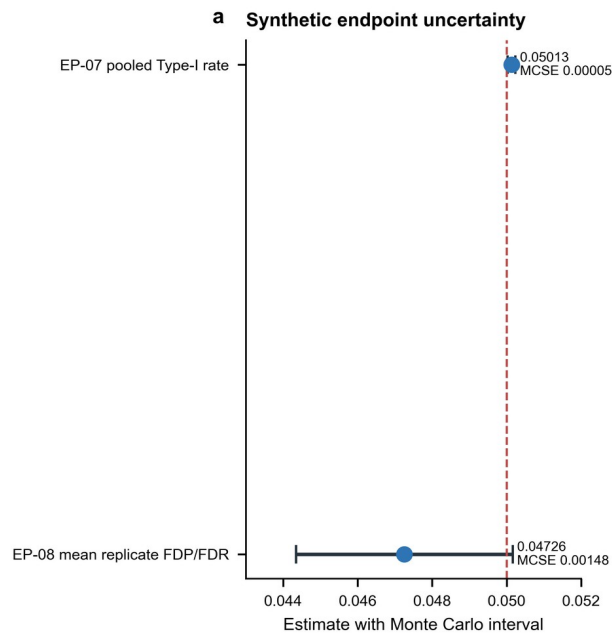


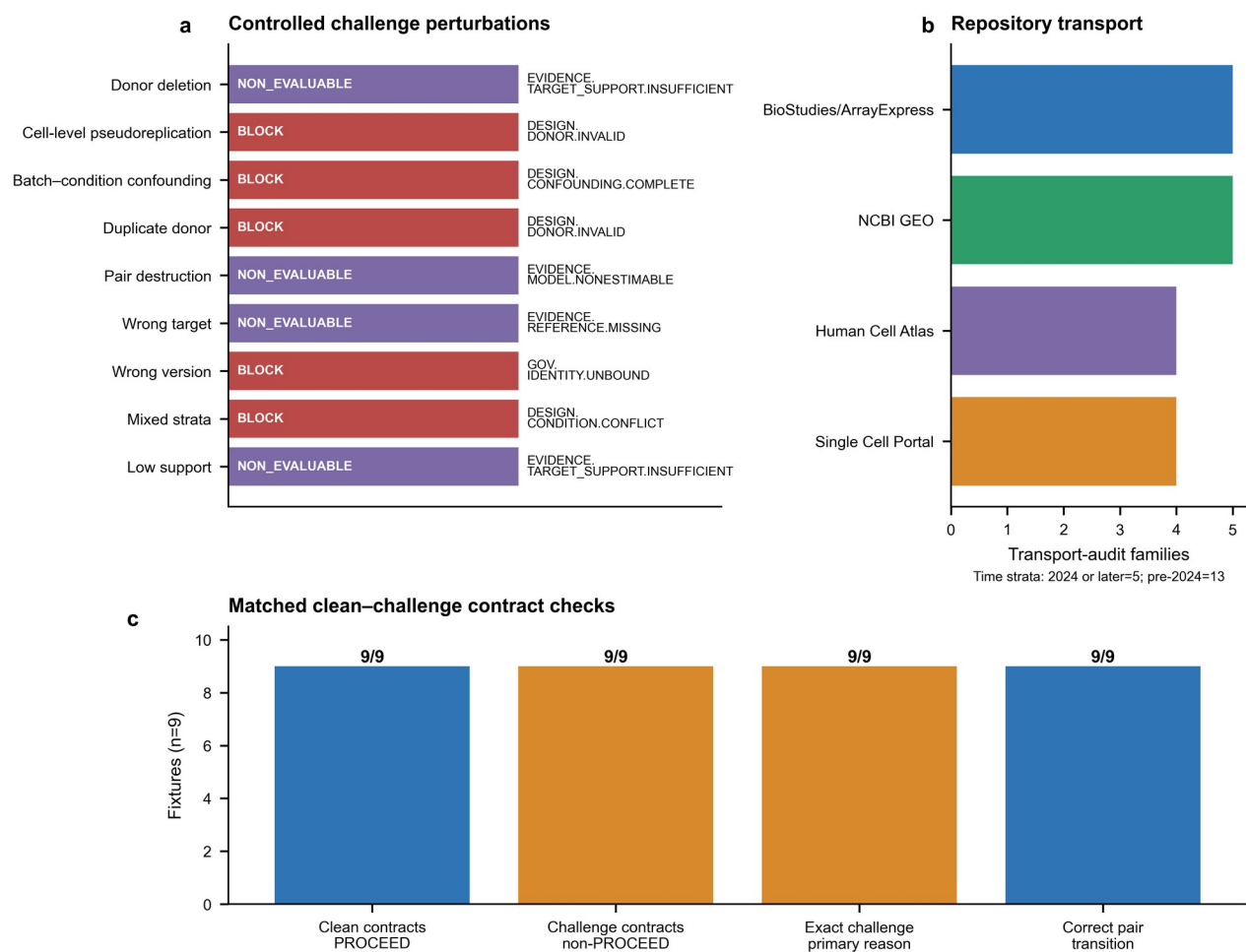
c Calibration-label composition

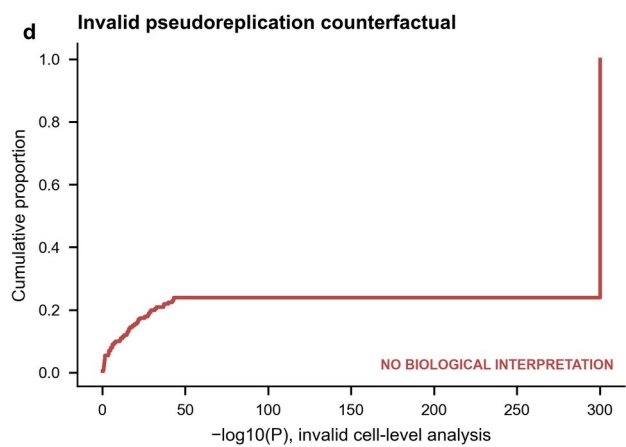
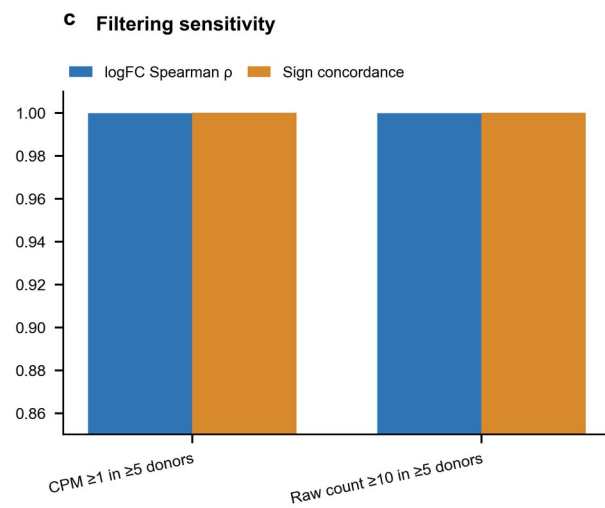
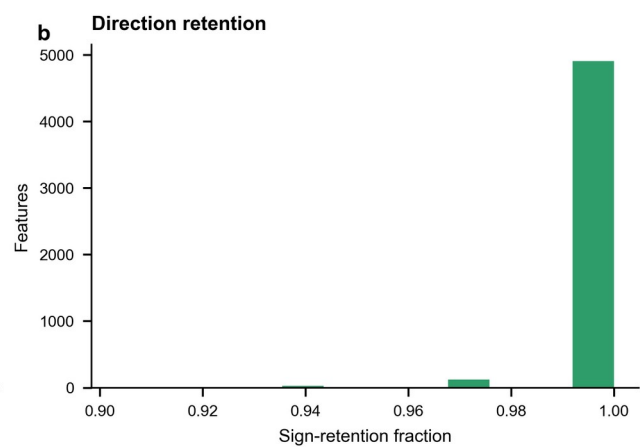
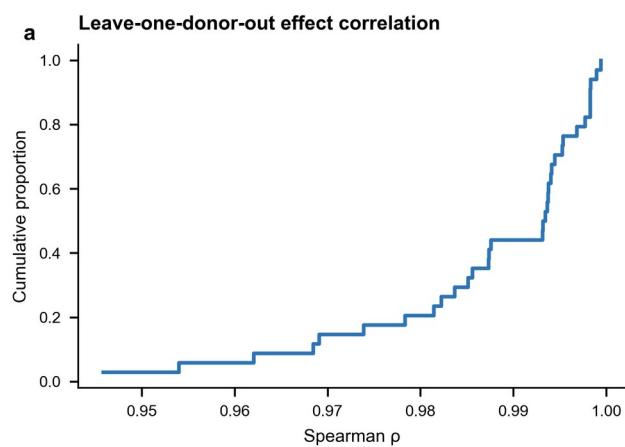


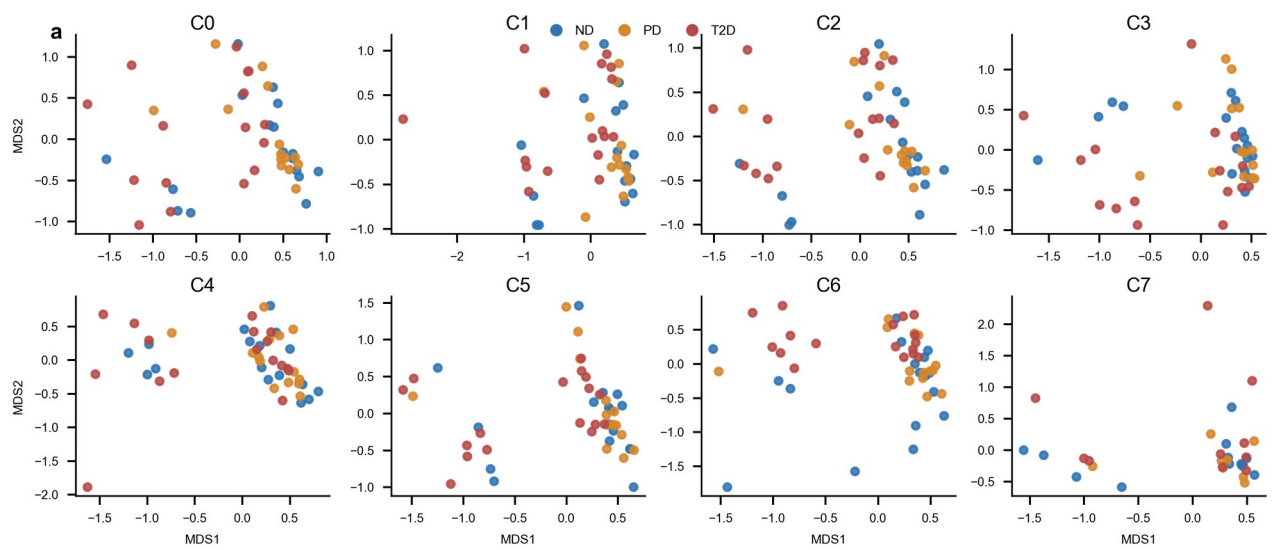
d Target-support floor











b Qualifying donors (≥ 20 cells)

	ND	PD	T2D
C0	17	14	16
C1	17	14	17
C2	17	14	17
C3	17	14	16
C4	17	14	17
C5	17	14	17
C6	17	14	17
C7	13	8	10

c Recovered beta-cell composition

